INCH-POUND

MIL-DTL-83723/67B
w/AMENDMENT 1
12 August 2019
SUPERSEDING
MIL-DTL-83723/67B
29 June 2009

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, ENVIRONMENT RESISTING, PLUG, THREADED COUPLING, PUSH-PULL, QUICK-DISCONNECT WITHOUT LANYARD, CRIMP SOCKET CONTACT, SERIES III, CLASSES A, G, M, R, T, W AND Z

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

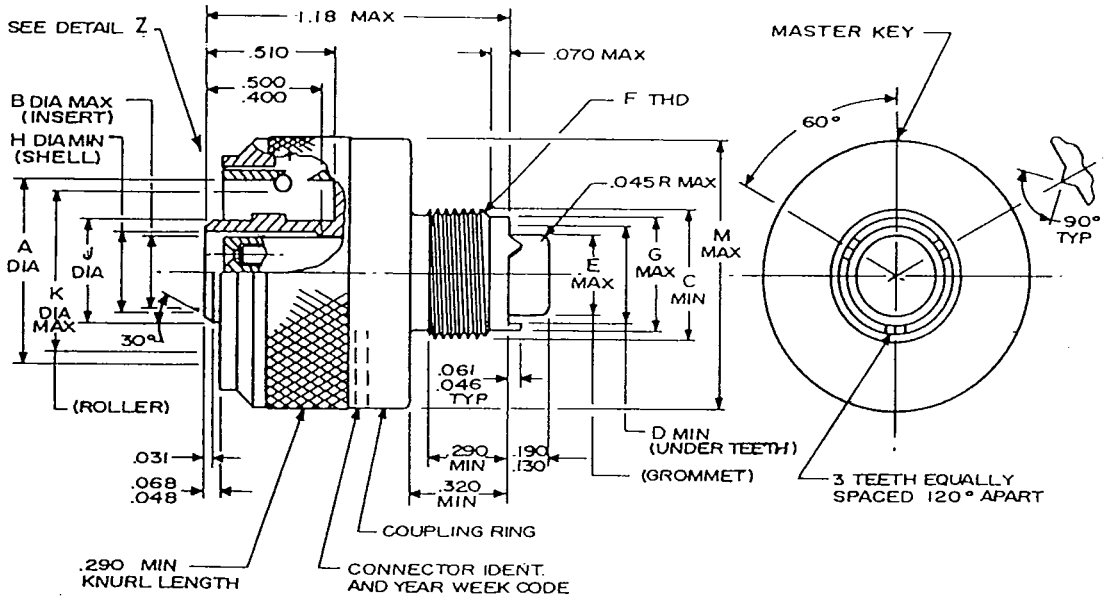
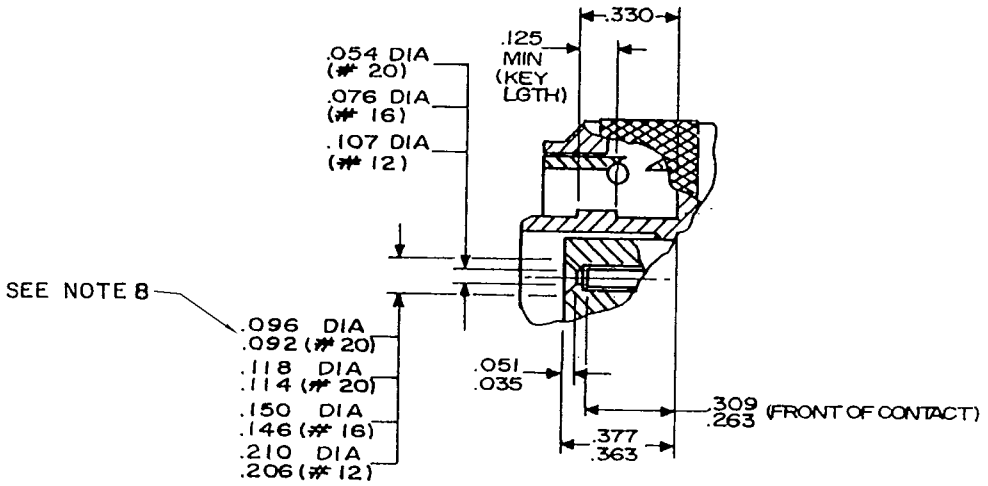


FIGURE 1. Dimensions and configurations.

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DETAIL Z

Shell size	A dia	B dia	C dia	D dia	E dia	F thread	G dia	H dia
8	.704 (17.88)	.290 (7.37)	.470 (11.94)	.370 (9.40)	.305 (7.75)	.500-20 UNF-2A	.437 (11.10)	.294 (7.47)
10	.829 (21.06)	.388 (9.86)	.600 (15.24)	.497 (12.62)	.405 (10.29)	.625-24 UNEF-2A	.572 (14.53)	.428 (10.87)
12	1.016 (25.81)	.558 (14.17)	.724 (18.39)	.613 (15.57)	.531 (13.49)	.750-20 UNEF-2A	.687 (17.45)	.598 (15.19)
14	1.079 (27.41)	.627 (15.93)	.849 (21.56)	.738 (18.75)	.665 (16.89)	.875-20 UNEF-2A	.812 (20.62)	.667 (16.94)
16	1.204 (30.58)	.772 (19.61)	.974 (24.74)	.863 (21.92)	.790 (20.07)	1.000-20 UNEF-2A	.937 (23.80)	.794 (20.17)
18	1.345 (34.16)	.860 (21.84)	1.030 (26.16)	.919 (23.34)	.869 (22.07)	1.0625-18 UNEF-2A	.992 (25.20)	.900 (22.86)
20	1.470 (37.34)	.985 (25.02)	1.154 (29.31)	1.044 (26.52)	.994 (25.25)	1.1875-18 UNEF-2A	1.117 (28.37)	1.025 (26.04)
22	1.595 (40.51)	1.110 (28.19)	1.279 (32.49)	1.169 (29.69)	1.119 (28.42)	1.3125-18 UNEF-2A	1.242 (31.55)	1.150 (29.21)
24	1.720 (43.69)	1.235 (31.37)	1.404 (35.66)	1.294 (32.87)	1.244 (31.60)	1.4375-18 UNEF-2A	1.367 (34.72)	1.275 (32.39)

FIGURE 1. Dimensions and configurations - Continued.

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Shell size	J	K	M
8	.405 (10.29) .396 (10.06)	.612 (15.54)	1.095 (27.81)
10	.526 (13.36) .517 (13.13)	.737 (18.72)	1.240 (31.50)
12	.696 (17.68) .687 (17.45)	.925 (23.50)	1.432 (36.37)
14	.765 (19.43) .756 (19.20)	.987 (25.07)	1.490 (37.85)
16	.892 (22.66) .883 (22.43)	1.112 (28.24)	1.711 (43.46)
18	.998 (25.35) .989 (25.12)	1.237 (31.42)	1.815 (46.10)
20	1.123 (28.52) 1.114 (28.30)	1.362 (34.59)	1.962 (49.83)
22	1.248 (31.70) 1.239 (31.47)	1.487 (37.77)	2.070 (52.58)
24	1.373 (34.87) 1.364 (34.65)	1.612 (40.94)	2.195 (55.75)

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.031	0.79	.363	9.22	.667	16.94	.992	25.20	1.294	32.87
.035	0.89	.367	9.32	.687	17.45	.994	25.25	1.313	33.35
.045	1.14	.370	9.40	.696	17.68	.998	25.35	1.345	34.16
.051	1.30	.377	9.58	.704	17.88	1.000	25. 40	1.362	34. 59
.054	1.37	.378	9.60	.705	17.91	1.016	25.81	1.364	34.65
.061	1.55	.388	9.86	.708	17.98	1.025	26.04	1.367	34.72
.068	1.73	.396	10.06	.724	18.39	1.044	26.52	1.373	34.87
.070	1.78	.400	10.16	.730	18.54	1.063	27.00	1.432	36.37
.076	1.93	.402	10.21	.737	18.72	1.079	27.41	1.438	36.53
.081	2.06	.428	10.87	.738	18.75	1.090	27.69	1.470	37.34
.087	2.21	.437	11.10	.750	19.05	1.095	27.81	1.487	37.77
.092	2.34	.465	11.81	.756	19.20	1.110	28.19	1.490	37.85
.096	2.44	.468	11.89	.765	19.43	1.112	28.24	1.595	40.51
.097	2.46	.470	11.94	.772	19.61	1.114	28.30	1.612	40.94
.107	2.72	.481	12.22	.790	20.07	1.117	28.37	1.711	43.46
.114	2.90	.497	12.62	.794	20.17	1.119	28.42	1.720	43.69
.118	3.00	.500	12.70	.812	20.62	1.123	28.52	1.815	46.10
.130	3.30	.510	12.95	.829	21.06	1.130	28.70	1.962	49.83
.146	3.71	.526	13.36	.849	21.56	1.150	29.21	2.070	52.58
.150	3.81	.531	13.49	.860	21.84	1.154	29.31	2.129	54.08

FIGURE 1. Dimensions and configurations - Continued.

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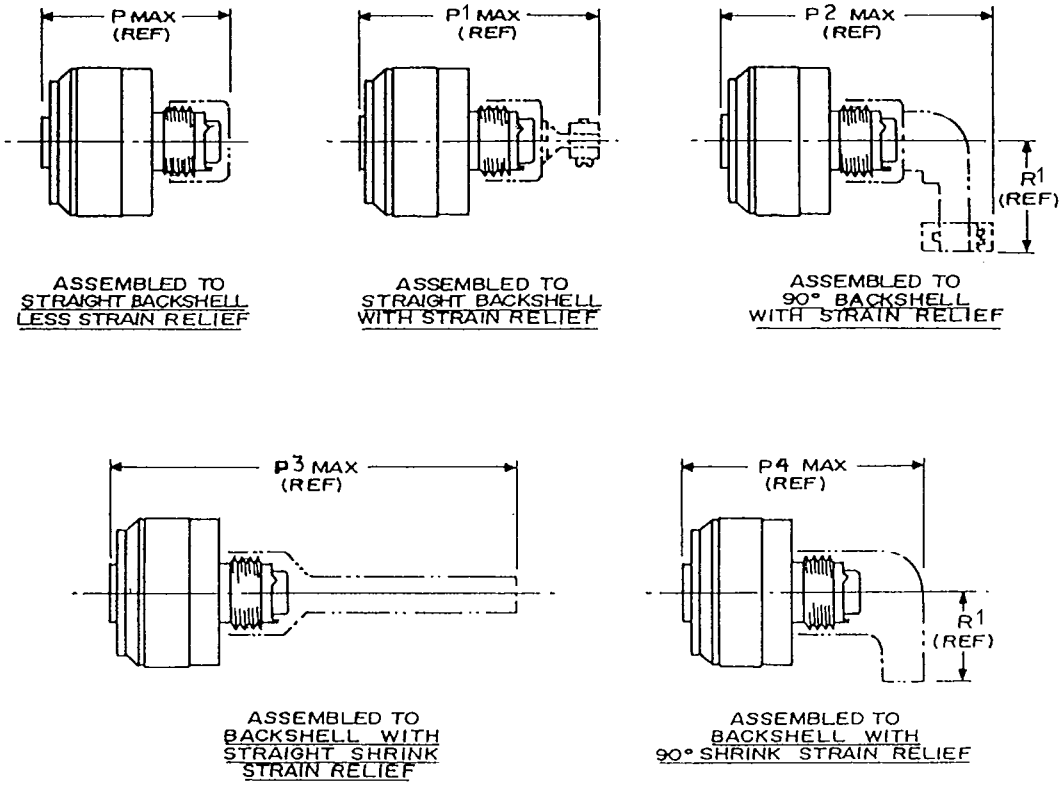
Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.190	4.38	.558	14.17	.863	21.92	1.169	29.69	2.195	55.75
.206	5.23	.572	14.53	.869	22.07	1.180	29.97	2.353	59.77
.210	5.33	.580	14.73	.875	22.23	1.188	30.18		
.222	5.64	.583	14.81	.883	22.43	1.204	30.58		
.225	5.72	.598	15.19	.892	22.66	1.224	31.09		
.263	6.68	.600	15.24	.898	22.81	1.235	31.37		
.282	7.16	.612	15.54	.900	22.86	1.237	31.42		
.285	7.24	.613	15.57	.919	23.34	1.239	31.47		
.290	7.37	.625	15.88	.925	23.50	1.240	31.50		
.294	7.47	.627	15.93	.937	23.80	1.242	31.55		
.305	7.75	.643	16.33	.975	24.77	1.248	31.70		
.309	7.85	.646	16.41	.985	25.02	1.275	32.39		
.320	8.13	.662	16.89	.987	25.07	1.279	32.49		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .010 (0.25) for two place decimals, ± .005 (.13 mm) for three place decimals, and ± 1 **°** and ± 30’ for angular dimensions.
4. The dimensions showing the relationship between the coupling ring and shell apply only when the coupling ring is in the full forward position.
5. The quick-disconnect connector plug is intended to be used with series III threaded coupling receptacles MIL-DTL-83723/82 and MIL-DTL-83723/84 with the adapter MIL-DTL-83723/70 permanently attached to the receptacle.
6. Coupling of the quick-disconnect plug connector is to be accomplished by a straight axial push with no cocking of the coupling mechanism.
7. Uncoupling of the quick-disconnect plug connector is to be accomplished by a straight axial pull on the outer coupling ring.
8. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations - Continued.

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Shell size	P	P1	P2	P3	P4	R	R1
8	1.457 (37.01)	1.869 (47.47)	2.018 (51.26)	5.838 (148.29)	1.848 (46.94)	.830 (21.08)	4.585 (116.46)
10	1.457 (37.01)	1.979 (50.27)	2.100 (53.34)	6.528 (165.81)	1.888 (47.96)	.880 (22.35)	5.390 (136.91)
12	1.457 (37.01)	2.124 (53.95)	2.293 (58.24)	7.348 (186.64)	1.928 (48.97)	.950 (24.13)	6.330 (160.78)
14	1.457 (37.01)	2.124 (53.95)	2.353 (59.77)	7.348 (186.64)	1.928 (48.97)	1.010 (25.65)	6.330 (160.78)
16	1.457 (37.01)	2.229 (56.62)	2.502 (63.55)	8.068 (204.93)	2.028 (51.51)	1.070 (27.18)	6.710 (170.43)
18	1.457 (37.01)	2.474 (62.84)	2.614 (66.40)	8.068 (204.93)	2.028 (51.51)	1.130 (28.70)	6.710 (170.43)
20	1.457 (37.01)	2.604 (66.14)	2.739 (69.57)	9.068 (230.33)	2.148 (54.56)	1.190 (30.23)	7.740 (196.60)
22	1.457 (37.01)	2.729 (69.32)	2.864 (72.75)	9.068 (230.33)	2.148 (54.56)	1.260 (32.00)	7.740 (196.60)
24	1.457 (37.01)	2.854 (72.49)	2.989 (75.92)	9.068 (230.33)	2.228 (56.59)	1.320 (33.53)	8.030 (203.96)

FIGURE 2. Plug and backshell assembly, dimensions and configurations.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only. Metric equivalents are in parentheses.
3. Unless otherwise specified, tolerances are ± .010 (0.25) for two place decimals, ± .005 (0.13) for three place decimals, and ± 1° and ± 30' for angular dimensions.

FIGURE 2. Plug and backshell assembly, dimensions and configurations – Continued.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

For insert arrangements, see MIL-STD-1554 and table I.

TABLE I. Inserts requiring reduced diameters for lead-in chamfers on contact cavities.

Shell size	Insert arrangements	Contact cavities
8	03	1, 2, 3
10	06	1, 2, 3, 4, 5, 6

For connector accessories: See SAE-AS85049 and MIL-DTL-83723/16.

Unless otherwise specified in the purchase order, the following components shall be furnished with the connector:

a. Contacts in accordance with SAE-AS39029 shall be furnished with the connector. Spare contacts are required for each contact size. The total quantity of spare contacts shall be 10 percent or 1, whichever is greater, of the total number of each size contact required.

b. Grommet seal plugs in accordance with MS27488 shall be furnished with the connector. Total quantity shall be 15 percent of each contact size or 3, whichever is greater.

c. Contact extraction tools in accordance with SAE-AS81969 shall be included in the connector package.

Connector assembly shall include a backshell (see SAE-AS85049 and MIL-DTL-83723/16) to ensure compliance with moisture sealing requirements.

The coupling mechanism of the connector plug shall withstand a minimum tensile load of 100 lb (45.36 kg) minimum. Load shall be applied to rear accessory hardware in a straight axial direction.

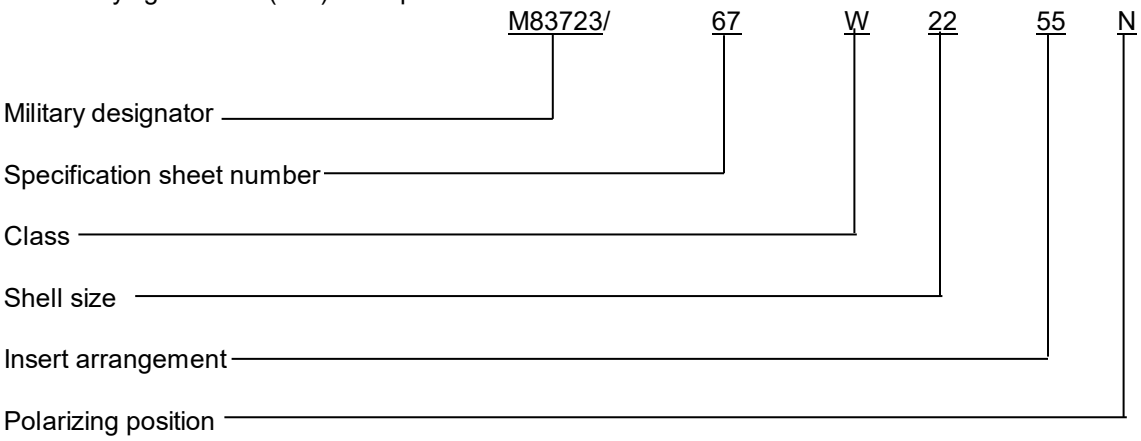
The quick-disconnect connector plug shall meet the vibration and shock requirements in accordance with MIL-DTL-83723 and the coupling mechanism shall maintain the plug and receptacle in a locked condition.

The tensile force required to unlock the coupling mechanism shall be 5 lb (2.27 kg) minimum and 20 lb (9.07 kg) maximum.

The locking mechanism shall meet the performance requirements of MIL-DTL-83723.

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Part or Identifying Number (PIN) example:



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Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-STD-1554
MS27488
MIL-DTL-83723/16
MIL-DTL-83723/70
MIL-DTL-83723/82
MIL-DTL-83723/84
SAE-AS39029
SAE-AS81969
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2019-104)

Review activity:
Air Force – 70

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